Exercise 3

Point of Sale System: it all the hardware and software that allows a business to sell their product. E.g in a restaurant; its the hardware and software that enables the guest to order, the software and hardware that allows the kitchen to see the order, the one that allows the guest to pay. all inclusive of hardware and software.

Our goal today is the specification of a Point of Sale (POS) system for a gas station chain.

• Functions per Point of Sale ◦ Listing the fuel prices of a petrol station ◦ Changing fuel prices ◦ Adding goods ◦ …

• Functions per petrol station chain / petrol station ◦ Adding a gas station ◦ Deleting a gas station

Task 1 – Test Objectives and Quality Attributes:

Define test objectives for your test project. Provide a justification your decision:

Test objectives, the user have a clear vision on the entrance of the station and the exit → this way we ensure usability and eliminate confusion for the user.

The user can see accurate up to date gas prices→ preventing overcharging or undercharging and maintaining good reputation.

The user can pay using their debit card→ the system can handle most common payment method.

The user can buy other goods and pay for them using debit card→ shows that the system can handle other non-gas related payment which will be profitable for business and the user.

Define the three most important quality attributes with the highest priority and justify your decision:

Reliability: the system should work at anytime, the failure of the paying system at anytime costs us money. In case the prices are not shown or the system shows higher prices the user will be then overcharged.

Usability: the process should be easy and intuitive for the user, everything should be where the user expects it to be.

Maintainability: the system should be extendable, adding more goods will give the costumer more options and us more revenue. It will be easier to fix bugs that will save the company money and save the costumer time.

What constructive/organizational quality assurance measures did you take in the run up to the acceptance test to ensure smooth acceptance and high test object quality? List f ive examples and briefly explain why the measures led to higher quality!:

Preparation of Detailed Requirements and Acceptance Criteria

- Measure: Documented all functional and non-functional requirements clearly, including expected behavior for fuel pricing, payments, and goods management.
- Why it improves quality: Provides a clear benchmark for the acceptance test, reducing misunderstandings and ensuring the system meets business needs.

Internal System Testing (Alpha Testing)

- Measure: Conducted internal testing by developers and staff to identify and fix obvious bugs before users test the system.
- Why it improves quality: Minimizes critical failures during acceptance testing and increases confidence that the system is stable.

Creation of Realistic Test Scenarios and Data

- Measure: Prepared test cases simulating real gas station operations, including multiple fuel types, payments, and goods transactions.
- Why it improves quality: Ensures users can test realistic situations, revealing usability issues and functional gaps before live deployment.

Training for Test Participants

- Measure: Provided short instruction sessions or guides to users who will perform the UAT.
- Why it improves quality: Ensures users can focus on testing the system rather than figuring out how it works, leading to more accurate and meaningful feedback.

Environment Setup and Data Verification

- Measure: Ensured the POS environment mirrors real operational conditions (e.g., correct fuel prices, inventory data, payment system integration).
- Why it improves quality: Prevents false errors caused by incorrect test setup and allows the acceptance test to evaluate the system realistically.

The following problems are observed during the acceptance test. Which of the problems represent failures, which represent defects? Why?

a) The system reacts with delay when adding a new product

b) The corporate identity color scheme is not applied consistently

c) Displayed fuel prices are outdated

Failure: the system reacts with delay when adding a new product, display of fuel prices are outdated → A failure occurs when the system does not perform as expected during operation the user experiences a problem. A failure is always visible to the user

Defect: the corporate identity color scheme is not applied consistently → A defect (also called a "bug") is a flaw, imperfection, or inconsistency in the system that violates requirements, specifications, or design standards. Not visible for the user, does not affect the behavior of the application.

<u>Task 2 – Errors:</u>

What are potential reasons for human made errors? Brainstorm as a team. List at least 5 reasons.

Lack of experience or training

Fatigue or stress

Miscommunication or unclear instructions

Distraction or inattention

Poor interface or confusing system design

<u>Task 3 – Traceability:</u>

What is traceability in testing? Why is maintenance of traceability important?

Traceability in testing is the ability to link requirements to test cases to ensure every requirement is verified.

Importance of maintaining traceability:

- Ensures all requirements are tested
- Helps identify gaps or missing tests

- Supports impact analysis when requirements change
- Improves accountability and documentation